

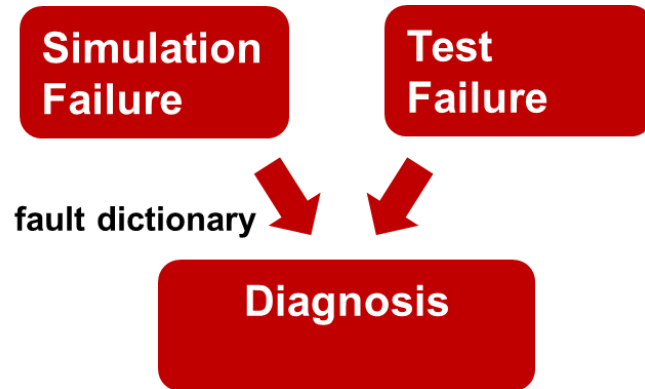
Diagnosis

- Introduction
- Logic Diagnosis
 - ◆ SSF diagnosis
 - * Static Cause-effect diagnosis
 - * Dynamic Cause-effect diagnosis
 - * Effect-cause diagnosis
 - Path Tracing: CPT, STAR
 - Pruning
 - Ranking
 - ◆ Unmodeled / multiple fault diagnosis
- Scan Chain Diagnosis
- Failure Analysis
- Conclusions

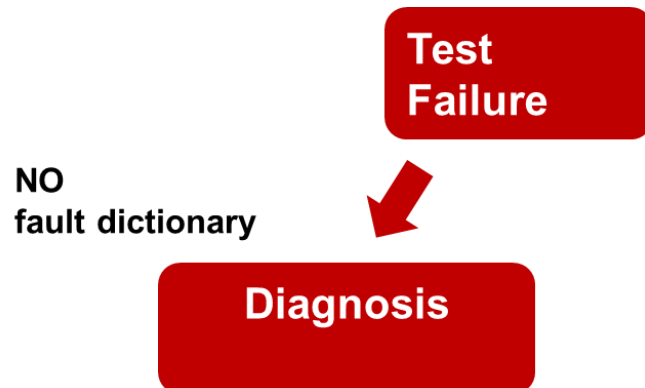


Review

cause-effect diagnosis



effect-cause diagnosis



Typical Effect-Cause Diagnosis Flow

test patterns, ckt

test failures

Path-Tracing

CPT, STAR

candidate faults

Pruning

Set operation, Layout, Timing ...

suspect faults

Ranking

Fault Simulation

ranked suspect faults

* This flow can be
modified in different
diagnosis tools

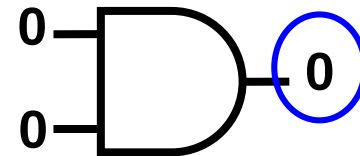
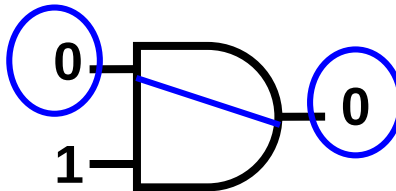
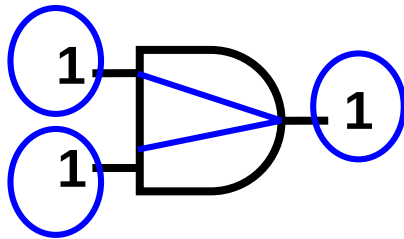
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Review: Critical Path (see CH 5)

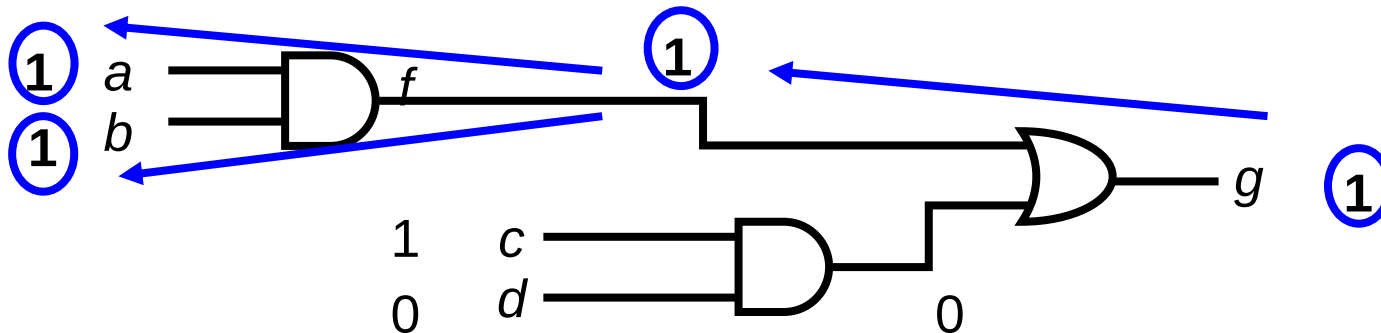
- x is **Critical Signal**
 - ♦ x 's value change causes some **primary output** values to change
- Example: when gate **output** is critical 
 - ♦ (1) If **all gate inputs are non-controlling**, they are all critical
 - ♦ (2) If **only one gate input is controlling**, it is critical
 - ♦ (3) Otherwise, no input is critical



critical signals are circled

Review: Critical Path Tracing (see CH 5)

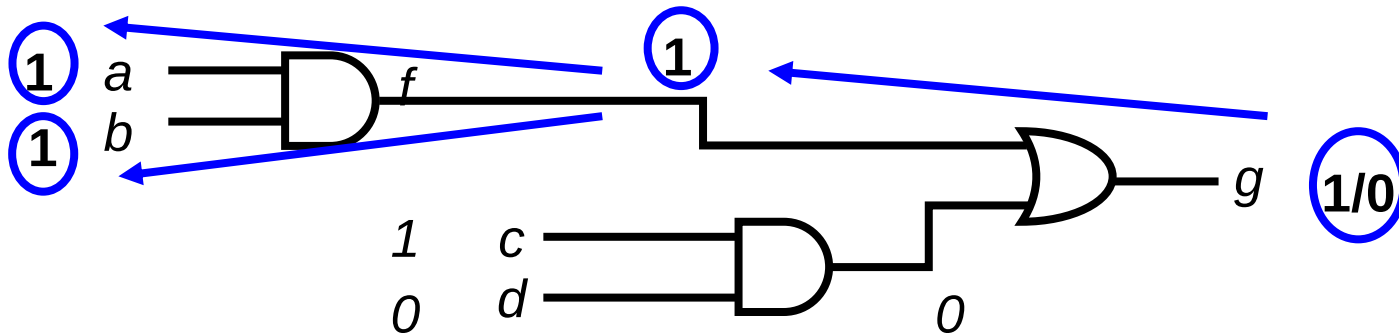
- **CPT** : trace critical signals from output
- CPT is simple in **fanout-free cone (FFC)**
 - ♦ **Linear time** algorithm. **No fault simulation** needed
- Example: Critical signals: *a, b, f, g*.



```
CPT (O) /*O is a node*/
1. foreach gate input g of O
2.   if (g is critical signal) then
3.     CPT (g) ;
4.   else return;
```

CPT for Diagnosis

- CPT from failing output
 - ♦ Critical signals are **candidate fault sites**
 - ♦ Stuck values are opposite to their good logic values
- Example: CPT from failing output g
 - ♦ 4 candidate faults: g SA0, f SA0, a SA0, b SA0



THEOREM:

Assume there is only one SSF in FFC.

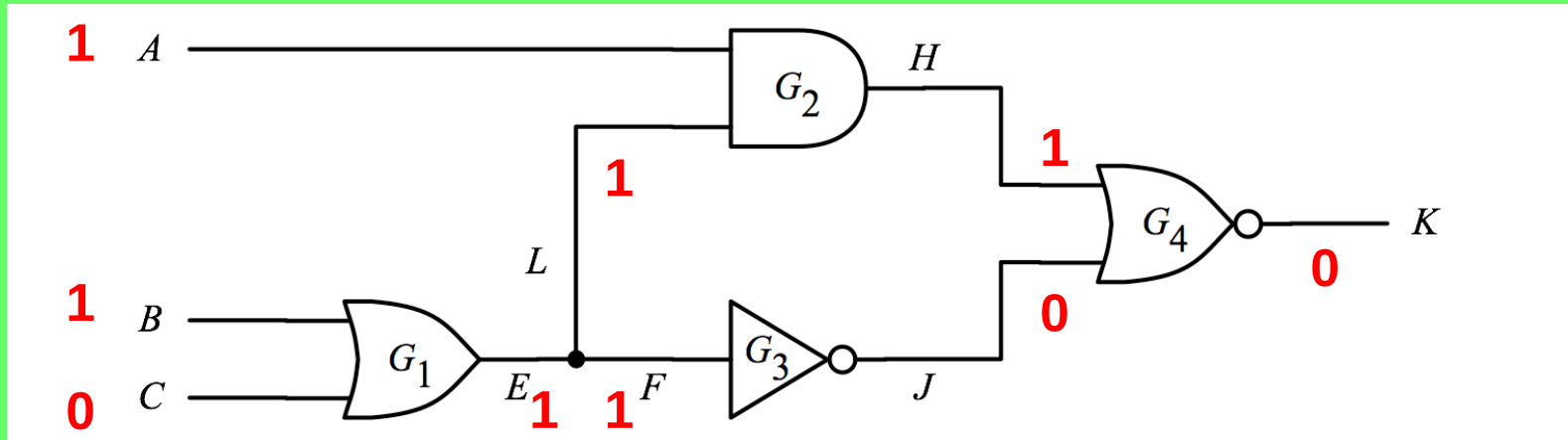
The fault must be one of candidate faults using CPT.

Quiz

Q: Suppose a CUT fails this test pattern.

(1) Use CPT to find candidate faults. (2) How do you know fanout stem E is critical or not?

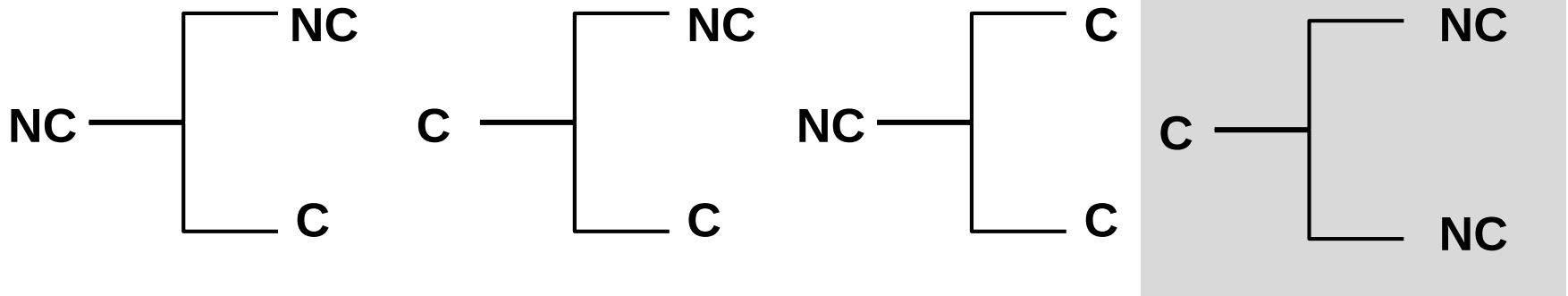
A:



CPT Cannot Decide Criticality of Fanout Stem

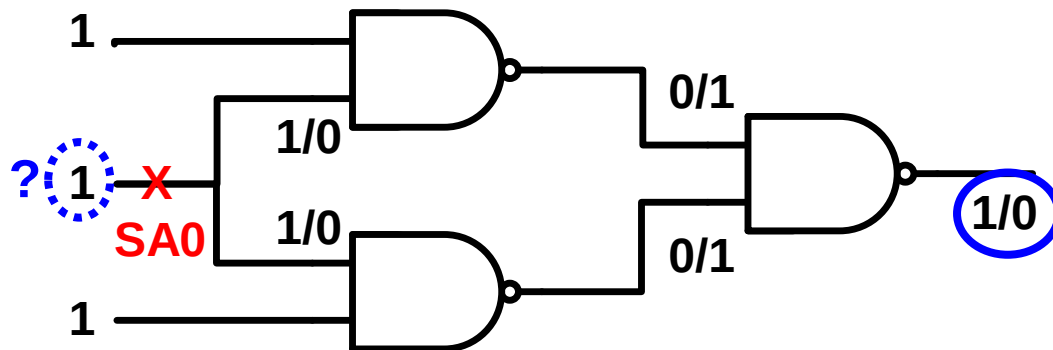
Four Cases for Fanout

- All four cases are possible



- Case 4 example: *multiple path sensitization*

- ♦ No branch is critical but stem is critical. How to trace?



Case 4 Very Bad for Diagnosis

Diagnosis

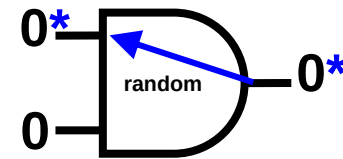
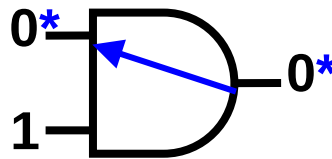
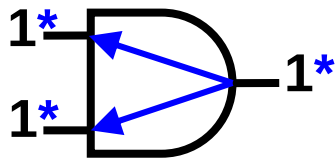
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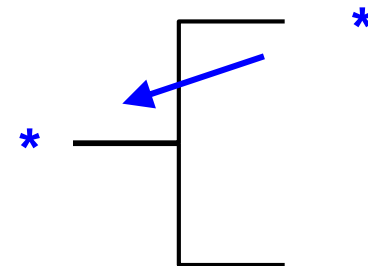
STAR Can Handle Fanouts

STAR Tracing [Akers 90]

- A **starred signal** means that it can be a candidate fault site
- When a gate output is starred
 - ♦ (1) If all gate inputs are non-controlling, they are all starred
 - ♦ (2) If only one gate input is controlling, it is starred
 - ♦ (3) Otherwise, randomly select one to be starred

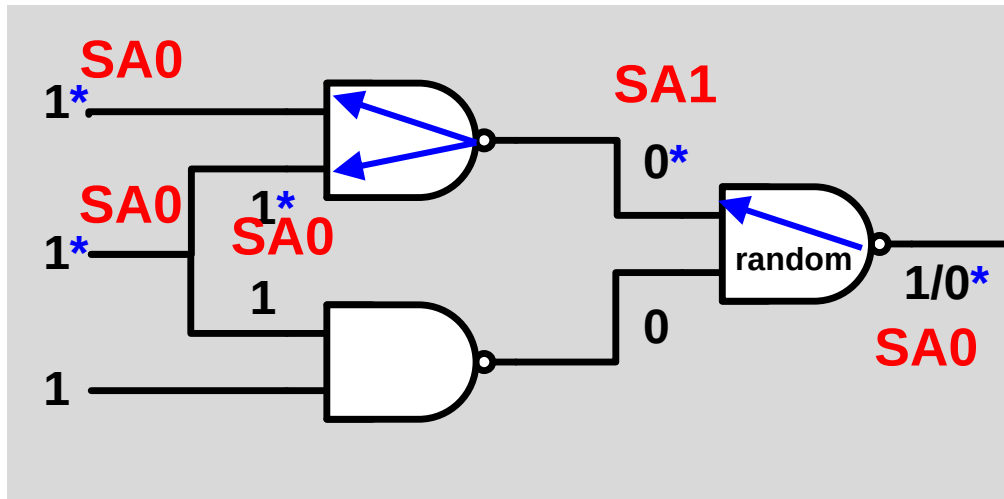


- When **any fanout branch** is starred, its fanout stem is starred
 - ♦ otherwise, fanout stem is NOT starred



STAR Tracing for Diagnosis

- Run StarTracing() from each failing output
 - ♦ Starred signals are candidate fault sites
 - ♦ Stuck-at values are opposite to its good logic value
- Example: 5 candidate faults

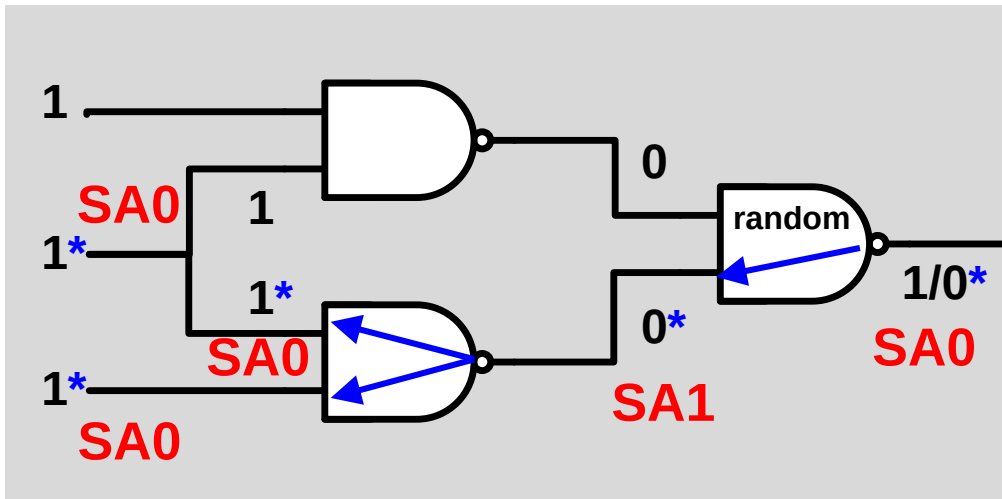


StarTracing (O)

1. **foreach** gate input g of O
2. **if** (g is starred) **then**
3. StarTracing (g) ;
4. **else return;**

Another STAR Tracing

- STAR tracing can be performed **multiple times** for a pattern
 - ♦ Each time can result in **different** candidate faults
- Example: 5 (different) candidate faults
 - ♦ Fanout stem is still one of candidates



StarTracing (O)

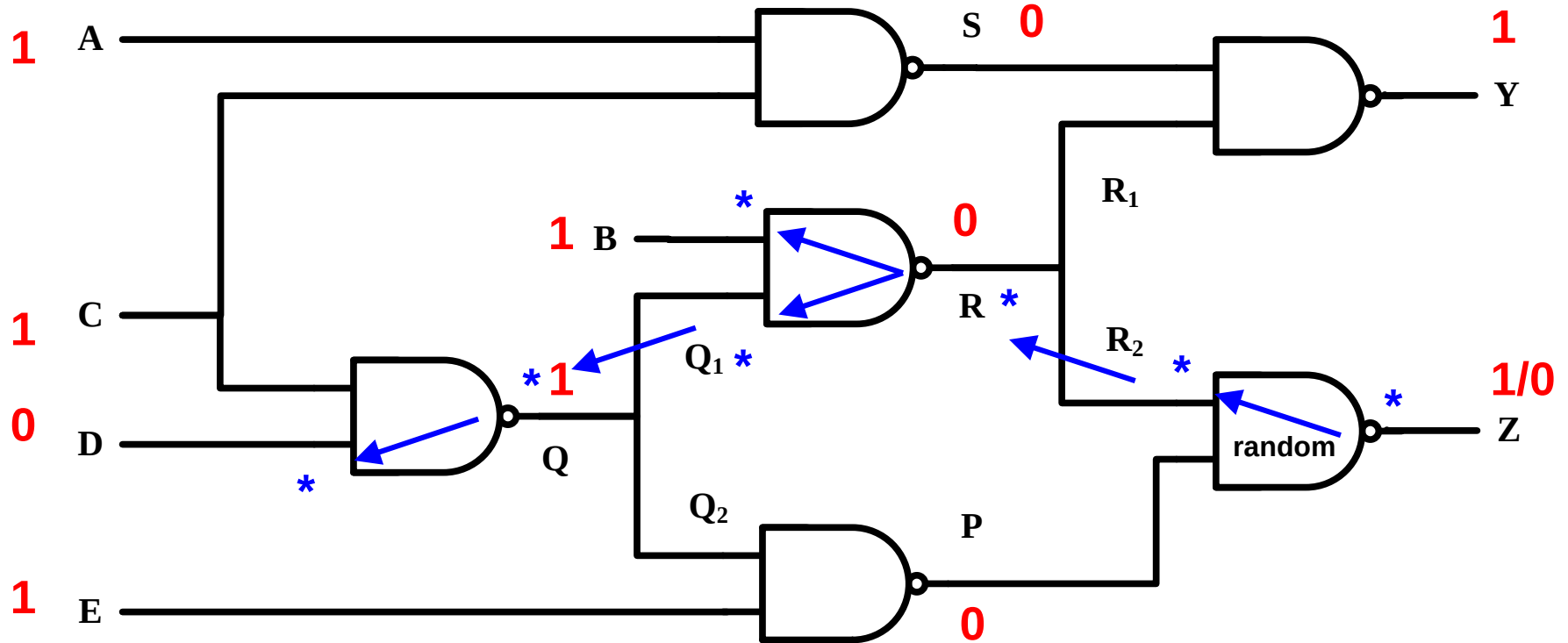
1. **foreach** gate input g of O
2. **if** (g is starred) **then**
3. StarTracing (g) ;
4. **else return**;

THEOREM: (to be proved in FFT)

Assume there is only one SSF in CUT. The fault must be one of candidate faults using STAR Tracing.

STAR Tracing Example (pattern 1/2)

- Example: STAR Tracing from failing output Z
 - ◆ 7 candidate faults: Z SA0, R₂ SA1, R SA1, Q₁ SA0, Q SA0, B SA0, D SA1

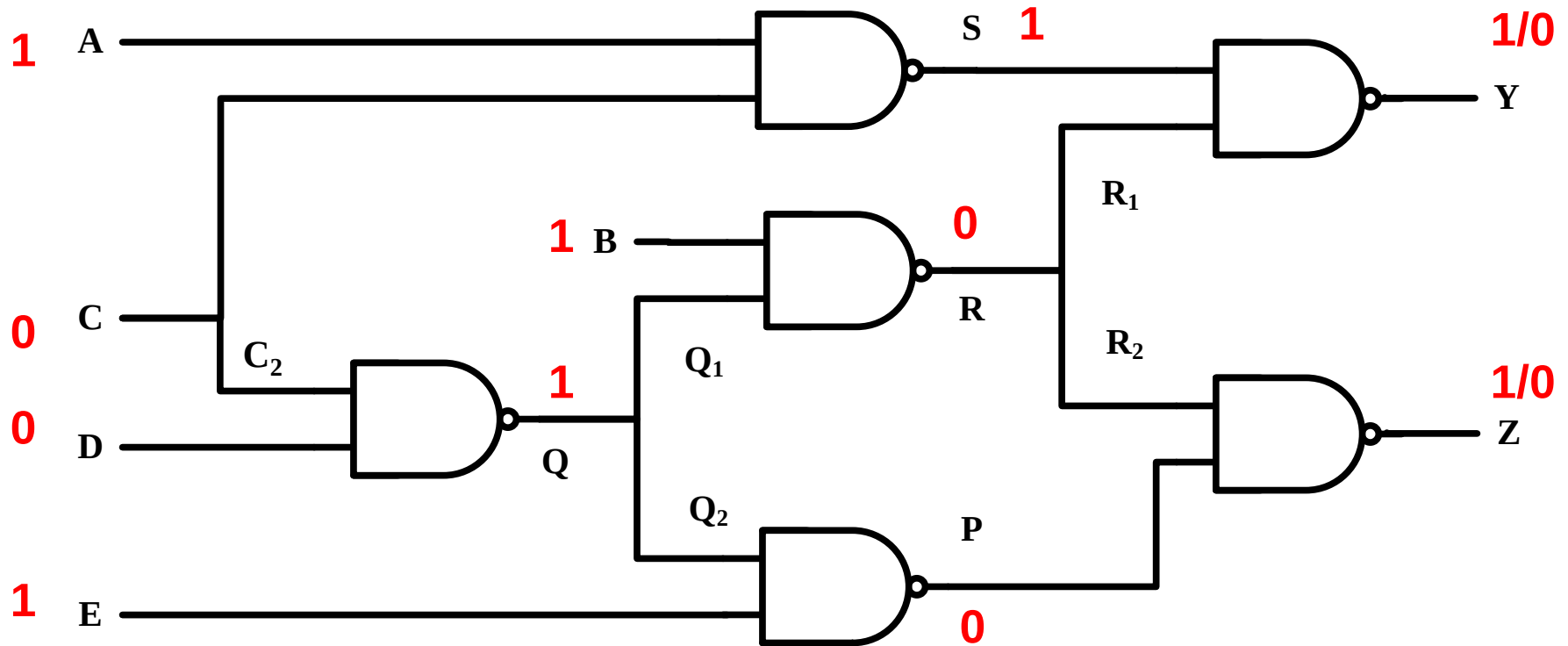


STAR Tracing Example (pattern 2/2)

Q: Apply another pattern. Y and Z are failing outputs.

(1) Which signals are starred? (2) What are candidate faults?

(Suppose we randomly select P and C)



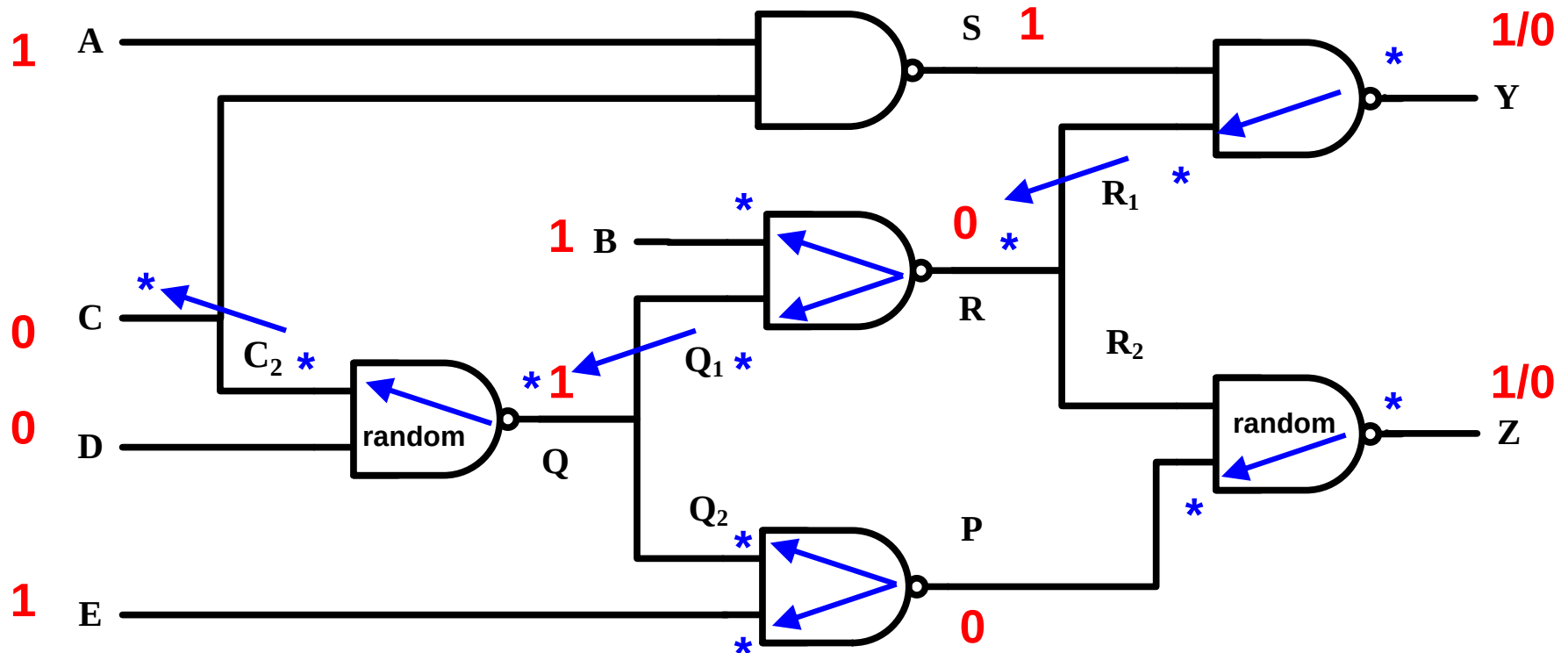
ANSWER

Q: Apply another pattern. Y and Z are failing outputs.

(1) Which signals are starred? 12

(2) What are candidate faults?

Y SA0, Z SA0, R₁ SA1, R SA1, P SA1, B SA0, Q₁ SA0, Q₂ SA0, Q SA0, C₂ SA1, C SA1, E SA0



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candidate faults

Pruning

Set operation, Layout, Timing ...

suspect faults

Ranking

Fault Simulation

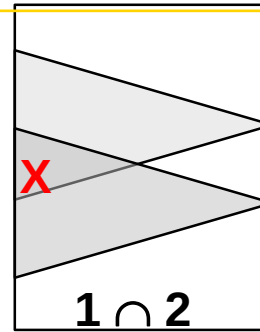
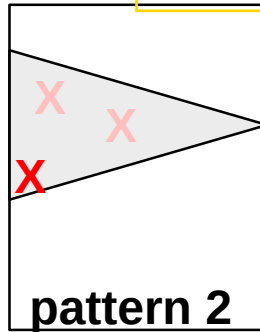
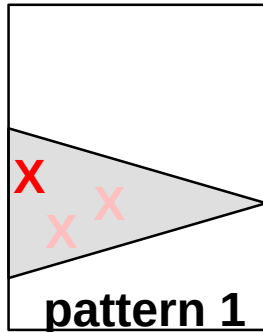
ranked suspect faults

Single Fault / Multiple Faults

(*not in exam)

- Single fault : take **intersection** of candidate faults

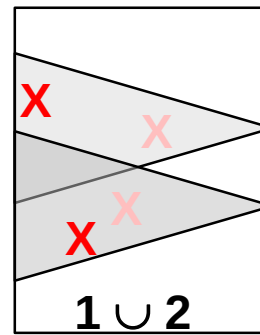
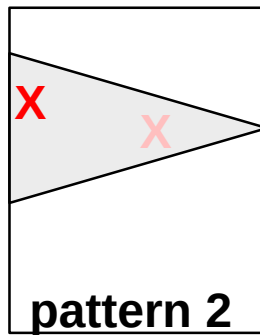
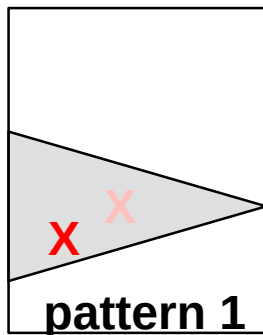
- ♦ guaranteed to be correct



X true fault
X false fault

- Multiple fault: take **union** of candidate faults

- ♦ may be incorrect : can miss some true faults (Why? FFT)



Example (Cont'd)

- After pattern 1

- 7 candidate faults:

Z SA0, R₂ SA1, R SA1, Q₁ SA0, Q SA0, B SA0, D SA1

- After pattern 2

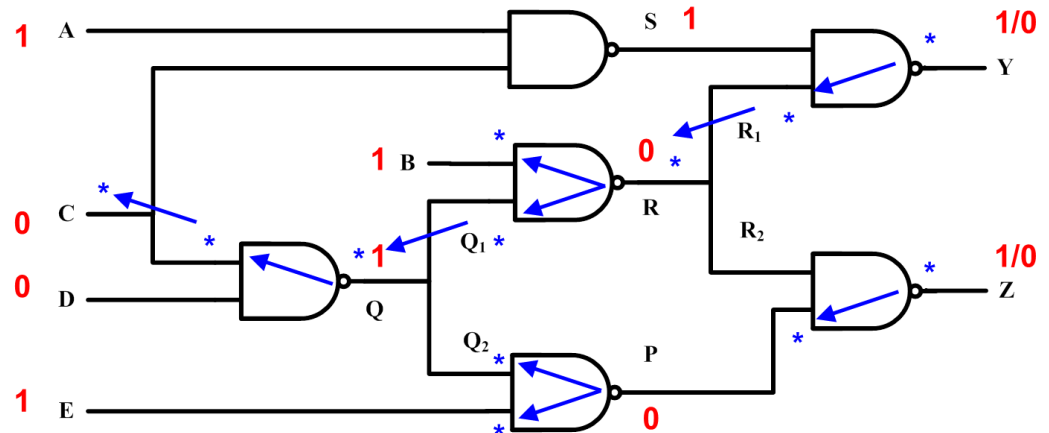
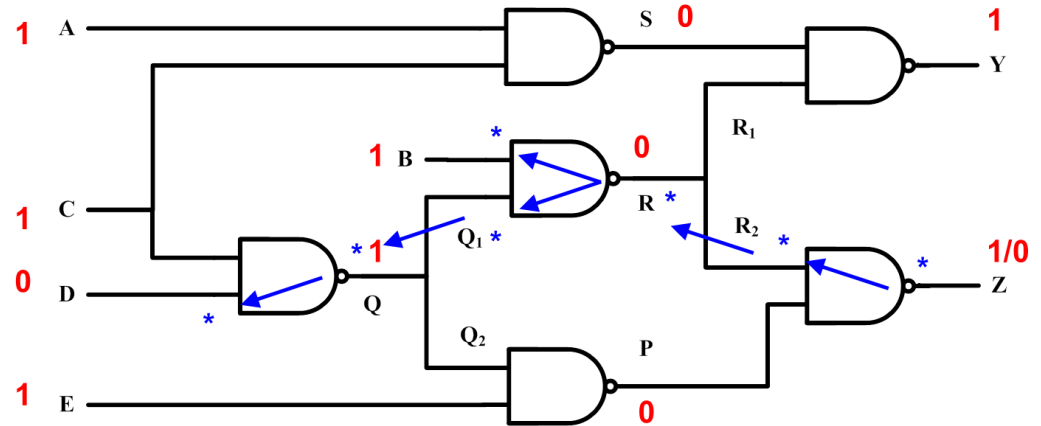
- 12 candidate faults:

Y SA0, Z SA0, R₁ SA1, R SA1, P SA1, B SA0, Q₁ SA0, Q₂ SA0, Q SA0, C₂ SA1, C SA1, E SA0

- After Intersection

- 5 suspect faults

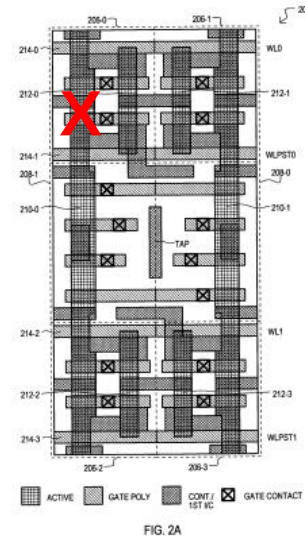
Z SA0, R SA1, Q₁ SA0, Q SA0, B SA0



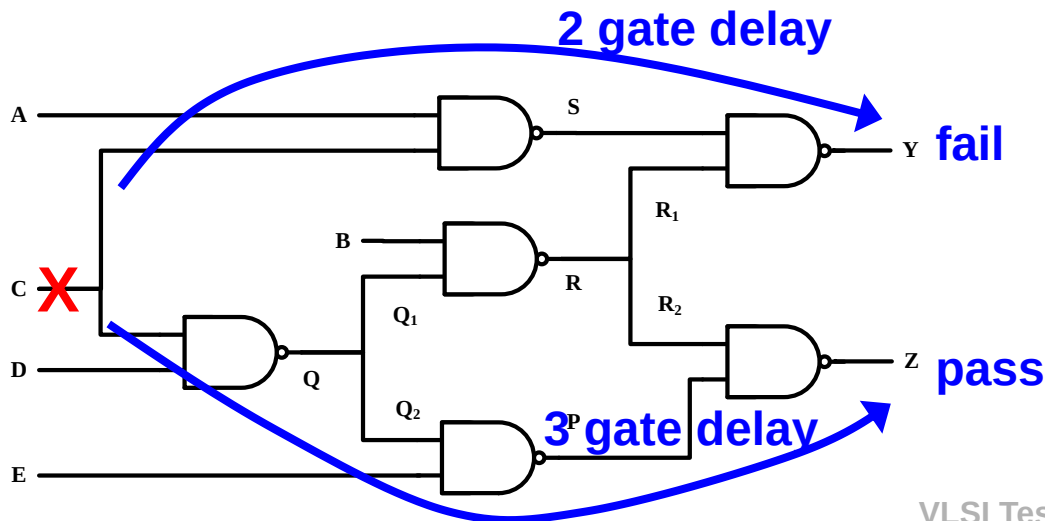
Other Pruning Techniques

(* not in exam)

- **Layout information** for bridging fault diagnosis
 - ♦ **Bridging pairs** must be neighbors on layout
 - ♦ If two faults are physically far from each other
 - * They can be pruned
- **Timing information** for delay fault diagnosis
 - ♦ If a fault fails short path but pass longer path
 - * It can be pruned



X



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Ranking

Fault Simulation

ranked suspect faults

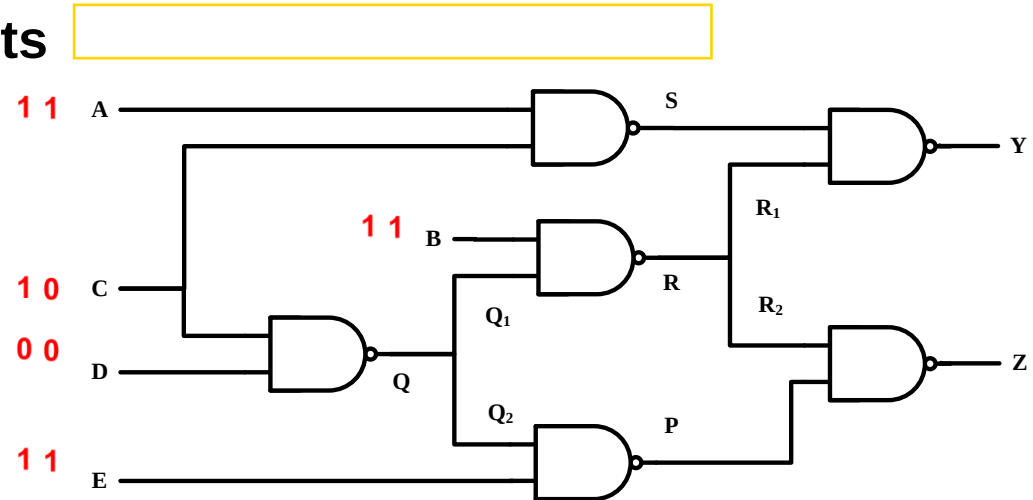
Ranking

- Fault simulate 5 suspect faults

♦ Z SA0, R SA1, Q₁ SA0, Q SA0, B SA0

- Q SA0 is ranked #1

$$\text{Diagnosis Score} = \frac{TFSF}{TFSP + \alpha \cdot TPSF + TFSF} \times 100\%$$



	faults	pattern1		pattern2		Diagnosis score			
		Y	Z	Y	Z	TFSF	TPSF	TFSP	score
1	Z SA0		X		X	2		1	0.67
2	R SA1			X		1		2	0.33
3	Q ₁ SA0		X	X		2		1	0.67
4	Q SA0		X	X	X	3			1.00
5	B SA0			X		1		2	0.33
	Test Failure		X	X	X				

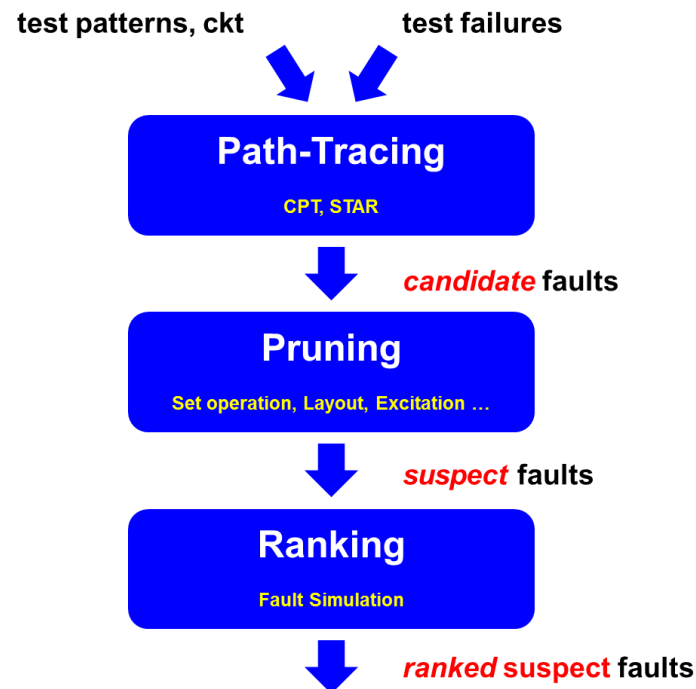
#2

#2

#1

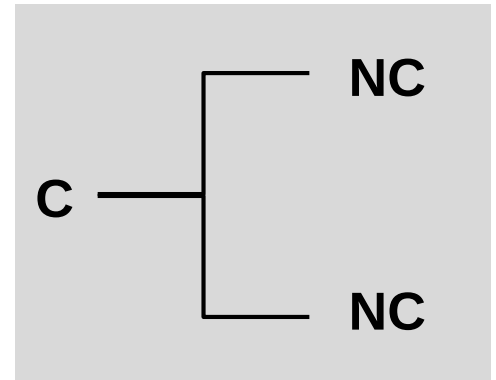
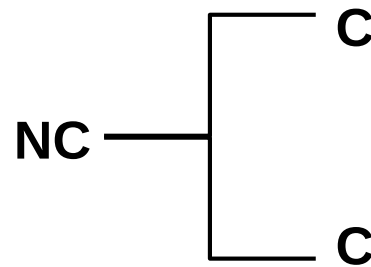
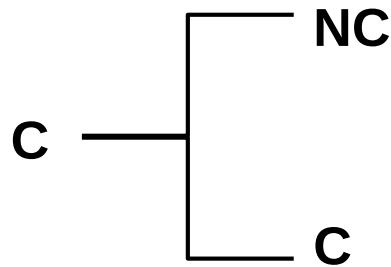
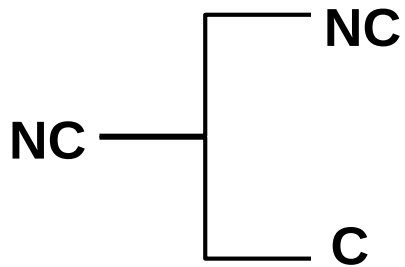
Summary

- Effect-Cause Diagnosis
 - ◆ No dictionary is needed
 - ◆ Most effective diagnosis nowadays
- Typical flow
 - ◆ Path tracing, Pruning, Ranking
- STAR tracing
 - ◆ Fast. No fault simulation needed.
 - ◆ Guaranteed correct if SSF
 - * Multiple path sensitization is OK



FFT #1

- Q: All four cases are possible
 - ♦ Please give examples for cases 1, 2, 3
 - ♦ Can we handle them by STAR tracing?

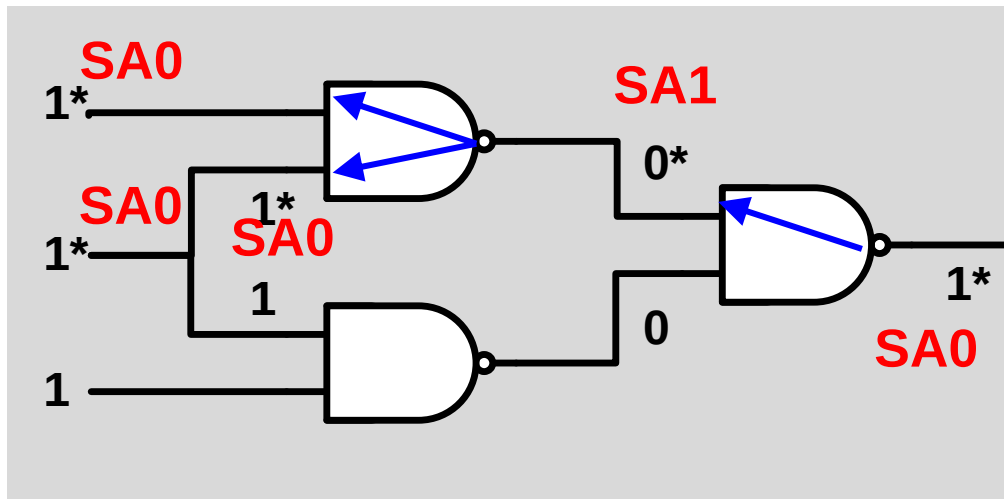


FFT #2

- Please prove the theorem

THEOREM:

Assume there is only one SSF in CUT. The fault must be one of candidate faults using STAR Tracing.



StarTracing (O)

1. **foreach** gate input g of O
2. **if** (g is starred) **then**
3. StarTracing (g) ;
4. **else return;**

FFT #3

- Multiple fault: take **union** of candidate faults
 - ♦ may be incorrect
 - * can miss some true suspect faults
- Q: Why we cannot guarantee to find all multiple faults using STAR tracing?

